

Workshop on Computational Fluid Dynamics (CFD)
with OpenFOAM
IMP-PAN Gdańsk

Tutorial #3: Compressible air Ejector
Supersonic Transient and Steady Flow

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Learning Objectives

By the end of this tutorial the student will be able to

1. **Navigate** OpenFOAM case directory structure
2. **Understand** basic OpenFOAM file formats
3. **Run** a simple transient simulation
4. **Analyze** basic results in terminal and **paraview**

1 Introduction and Theoretical Background

1.1 Case Overview

2 Case Setup

3 Part A: Density based explicit solver `rhoCentralFoam`

3.1 Running the Simulation

3.1.1 Mesh generation

3.1.2 Initialization and solver execution

3.2 Post-Processing in Terminal

3.3 Visualization in paraview

4 Part B: Density based implicit solver `HiSA`

4.1 Post-Processing in Terminal

4.2 Visualization with paraview

5 Student Exercises and Extensions

5.1 Basic exercises for Part A: VOF

5.2 Basic exercises for Part B: Euler-Euler

Disclaimer

This tutorial has been developed by **Robert Castilla**, of the Departament of Fluid Mechanics in the Universitat Politècnica de Catalunya, for the doctoral workshop at **IMPPAN Gdańsk** (July 2026). The author utilized **DeepSeek** for the initial definition of the document format. Further technical refinement and the development of advanced post-processing exercises were supported by **NotebookLM**. All the technical information for the **OpenFOAM v2512** environment and all physical interpretations were performed and approved by the author.